Investigating Bag-Level Imbalance and Sparse Attention in Histopathology Multiple Instance Learning

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***Abstract*—Whole-slide histopathology images contain rich diagnostic information, but their large size and the cost of detailed annotations make automated analysis challenging. Multiple instance learning (MIL) enables slide-level classification using weak supervision, offering a scalable approach to computational pathology. However, imbalanced class distributions and sparse diagnostic evidence within individual slides can limit model performance. Whether addressing these two challenges simultaneously improves classification remains unclear.**

This study investigates two modifications to attention-based multiple instance learning (ABMIL): inverse-frequency class weighting to address imbalance across slides and sparse attention aggregation to emphasize potentially informative patches within slides. Their individual and combined effects are evaluated on CAMELYON16, CAMELYON17, and BRACS using three random seeds, with additional experiments examining training-class prevalence, attention sparsity, and patient-independent validation.

The combined approach reduces mean test accuracy relative to standard ABMIL by 3.36, 11.33, and 6.59 percentage points across the three datasets, respectively. Sparse attention alone improves performance in selected settings but does not consistently generalize across datasets or evaluation protocols. hese findings suggest that independently motivated imbalance-handling strategies are not necessarily complementary. More broadly, they highlight the importance of evaluating interactions between optimization and evidence selection when developing reliable weakly supervised models for computational pathology.

***Index Terms*—Multiple instance learning, whole-slide images, gated attention, class imbalance, sparse aggregation, histopathology.**

I. Introduction

Whole-slide images (WSIs) are high-resolution digitized tissue sections. Their size makes exhaustive region annotation labor-intensive. Weakly supervised computational pathology therefore often uses slide-level labels rather than dense pixel-level annotations [1], [2]. Multiple instance learning (MIL) represents each slide as a bag of patch features and trains a classifier using its slide label. The conventional binary MIL assumption defines a positive bag as one containing at least one positive instance. This assumption does not directly describe every multiclass histological task.

Two sources of imbalance motivate this study. *Bag-level imbalance* refers to unequal class frequencies across training slides, a setting commonly addressed through reweighting or resampling [3]. *Within-bag evidence scarcity* refers to a small number of diagnostically informative patches within a slide, as can occur in metastasis detection [2], [4]. Gated ABMIL learns instance weights from bag labels [5], but does not explicitly correct class frequencies. Conversely, class-weighted training does not identify informative patches.

The research questions concern both the independent effects of the two interventions and their interaction. **H1** asks whether inverse-frequency class weighting improves minority-class performance. **H2** asks whether sparse attention improves bag-level classification when evidence competes with many other instances. **H3** asks whether combining the interventions adds benefit beyond either component. These are testable hypotheses, not predictions of universal improvement. H2 cannot be interpreted as a direct tumor-localization hypothesis without verified patch labels.

The study contributes a completed ABMIL training and evaluation pipeline, a controlled two-factor ablation on three datasets, and diagnostic analyses of prevalence, attention fraction, and patient independence. The primary experiment uses the original course splits. Later analyses are identified as exploratory because their plans followed the initial test observations.

II. Related Work

The multiple instance learning setting assigns a label to a bag without requiring labels for its constituent instances. In WSI classification, a bag represents a slide and its instances represent tissue patches. Mean pooling treats all patch features equally, which may dilute a small discriminative region among many unrelated patches. Element-wise max pooling can preserve strong responses but cannot learn which instances matter for a particular classification objective. These fixed aggregators do not use the slide label to adjust instance importance.

Attention-based multiple instance learning (ABMIL) addresses this limitation through a trainable, permutation-invariant aggregation operator. Ilse *et al.* [5] use a neural

attention module to assign normalized weights to the instances in each bag. Gated attention combines a hyperbolic tangent branch with a sigmoid branch, then pools features according to their learned scores. The attention and classification modules are optimized jointly from bag labels. This makes the contribution of an instance adaptive to its features and the training task. However, the scores are learned indirectly through bag classification. Attention is not automatically a faithful explanation of the decision [6]; a highly weighted patch is not necessarily a pathologically verified tumor region. Attention pooling also does not explicitly alter the contributions of minority-class bags to the supervised loss.

Clustering-constrained attention multiple instance learning (CLAM) extends attention pooling with additional instance-level training constraints [7]. Its motivation is the limited availability of slide annotations and the difficulty of identifying discriminative regions using only a slide classification objective. CLAM selects representative high- and low-attention instances and applies a clustering-based objective to encourage separation in the instance feature space. This introduces a second learning signal while retaining slide-level labels as the primary supervision. Unlike simply retaining high-attention patches during aggregation, CLAM changes the representation-learning objective. Its instance labels are weakly inferred rather than exhaustive pathologist annotations. Its design therefore illustrates one approach to exploiting limited supervision, but does not establish that highly attended patches are always tumor-positive.

Dual-stream multiple instance learning (DSMIL) addresses a related problem using a different aggregation structure [8]. An instance stream scores individual patches, while a bag stream relates the remaining instances to a critical, high-scoring instance. The resulting bag representation combines a salient local signal with information from related patches. The full DSMIL study also examines self-supervised feature extraction and multiscale representations. These upstream choices matter when comparing published systems: changes in feature quality cannot be attributed solely to the MIL aggregator. Relative to standard ABMIL, DSMIL explicitly models a reference-instance relationship rather than relying on independently produced attention logits followed by one weighted sum.

CLAM and DSMIL both recognize the difficulty of extracting decisive evidence from a large bag. Their mechanisms differ, however. CLAM adds representative-instance constraints to the learning objective; DSMIL changes how a critical instance informs bag aggregation. Neither mechanism is equivalent to hard top-fraction masking. A top-fraction operator excludes some features from the final pooled representation without adding patch labels or a new relational module. This distinction is important when interpreting the sparse-attention extension evaluated here.

TransMIL studies a limitation of treating patch embeddings as largely independent inputs to the pooling rule. Morphological and spatial correlations may carry information beyond the score assigned to any single patch. Shao *et al.* [9] use Transformer-based correlated instance modeling and positional encoding to incorporate these relationships. A Nyström approximation reduces the cost of attention over long instance sequences. Its design emphasizes that surrounding tissue can influence the meaning of an individual region. Consequently, removing low-scoring patches may discard context that a correlation-aware model could exploit. The present top-fraction experiment does not reproduce TransMIL or introduce positional reasoning into the classifier.

Double-tier feature distillation MIL (DTFD-MIL) focuses on another WSI-specific constraint: the number of labeled slides can be small even when each slide yields thousands of patches [10]. DTFD-MIL constructs pseudo-bags from partitions of a parent bag, then uses a two-tier distillation framework to combine their representations. Pseudo-bags increase the number of training units derived from each slide. They also introduce a label-related risk: a positive parent slide can produce a pseudo-bag with no positive instance. This risk connects limited bag-level supervision with sparse instance-level evidence. Unlike DTFD-MIL, the present study does not create pseudo-bags, transfer parent labels to sampled subsets, or train a second distillation tier.

These methods address complementary aspects of weakly supervised WSI classification. ABMIL learns instance importance; CLAM adds discriminative instance constraints; DSMIL uses a critical-instance relationship; TransMIL captures contextual dependencies; and DTFD-MIL restructures learning when labeled bags are scarce. They also differ in feature encoders, objectives, aggregation modules, and evaluation protocols. Performance reported in their original studies cannot be compared directly with results obtained from the course-provided frozen features and splits.

Two distinct forms of scarcity remain relevant to a minimal ABMIL pipeline. Reweighting is a standard response to class imbalance, although the effective number of examples may differ from the raw class count [3]. First, an underrepresented slide class contributes fewer training examples. A class-weighted objective can increase the gradient contribution of each available minority-class example, but cannot create new patient or slide diversity. Second, informative patches may constitute a small part of a large bag. Learned attention can prioritize such patches, but it may also attend to confounding structures or discard useful context when aggressively truncated. Sparse attention distributions have been explored with operators such as sparsemax [11]. The hard top-fraction mask evaluated here is a different operation, and no improvement is assumed. Importantly, the true informative-patch fraction is not observed in the present feature package. Sparse attention is therefore studied as an aggregation strategy, not as a verified tumor-patch selector.

A central empirical uncertainty is whether the two

TABLE I: Dataset sizes and the supplied course split.

Dataset	Classes	Train	Val.	Test
CAMELYON16	2	242	27	129
CAMELYON17	4	269	30	200
BRACS	7	389	62	86

interventions are complementary. Increasing the influence of rare-class slides could amplify useful supervisory signals. Restricting aggregation to high-scoring instances could also reduce dilution. Yet their combination could strengthen unreliable attention choices, especially when minority-class supervision is scarce. This interaction is not resolved by demonstrating either component in isolation. The research gap is deliberately bounded: under fixed course features, splits, and training budgets, the individual and joint effects of class weighting and sparse attention require direct evaluation. This is not a claim that broader MIL research has never studied imbalance or sparse evidence.

Two proposals follow from this distinction. The implemented proposal tests the separate and combined effects of loss weighting and sparse aggregation through a controlled ablation. A second proposal considers the stability of predictions when bag composition changes. Both address questions that can be posed without claiming access to patch-level tumor ground truth.

III. Data and Split Audit

The experiments use course-provided representations of three published pathology datasets: CAMELYON16 [2], CAMELYON17 [4], and BRACS [12]. Their original publications describe the source tasks; the course subsets and splits are specified below. The provided HDF5 files contain `feat`$\in \mathbb{R}^{N\times 384}$, `coords`$\in \mathbb{R}^{N\times 2}$, and a slide-label attribute [13]. The three datasets contain 15.53 GB of features in total. The feature extractor is upstream of this study. Its exact architecture, training cohort, and patch footprint cannot be verified from the supplied metadata. No encoder is retrained.

CAMELYON16 (C16) contains 398 bags and 4,641,464 instances. Training, validation, and test class counts are [142,100], [16,11], and [80,49]. CAMELYON17 (C17) contains 499 bags and 5,198,954 instances; its corresponding counts are [177,18,37,37], [20,2,4,4], and [121,16,18,45]. BRACS contains 537 bags and 5,400,641 instances. Its training class counts are [24,119,54,24,28,40,100]. The provided labels are retained: C17 is a four-class task, and BRACS is a seven-class task. Class indices are not assigned unverified names.

The original C17 train and validation sets share 27 patients, although its center-held-out test set has no patient overlap with either split. This weakens independent checkpoint selection. A supplementary split groups the non-test data by patient using the first fold of `StratifiedGroupKFold` (10 folds, seed 42). The resulting training and validation sizes are 269 and 30 slides; no patients overlap across splits. The grouped validation class counts are [20,1,5,4], making rare-class model selection unstable.

BRACS patient identifiers are not present in the supplied CSV or HDF5 attributes. The official dataset site describes additional metadata, but its mapping was not available locally [14]. Patient independence for BRACS and C16 is therefore not independently verified. No patient grouping is inferred from slide-number proximity.

Each bag is loaded lazily from HDF5. Valid instances are distinguished from padding by a boolean mask. Experiments use one bag per optimization step. Slide IDs, coordinates, and any additional unlabeled probability fields are excluded from classifier input. Coordinates are used only for visualization. Class frequencies are computed from training data.

IV. Method and Research Proposals

A. ABMIL Baseline

Let $\mathbf{x}_i \in \mathbb{R}^{384}$ be the feature of patch i. A trainable projection gives

$$\mathbf{h}_i = \mathrm{ReLU}(W_r \mathbf{x}_i + \mathbf{b}_r). \tag{1}$$

The 256-dimensional representation is scored by gated attention with 128-dimensional hidden branches:

$$s_i = \mathbf{w}^\top \left[\tanh(V\mathbf{h}_i) \odot \sigma(U\mathbf{h}_i)\right], \tag{2}$$

$$\alpha_i = \frac{\exp(s_i)}{\sum_{j\in\mathcal{V}} \exp(s_j)}, \qquad i \in \mathcal{V}, \tag{3}$$

where $\mathcal{V}$ is the set of valid patches. The bag representation and logits are

$$\mathbf{z} = \sum_{i\in\mathcal{V}} \alpha_i \mathbf{h}_i, \qquad \boldsymbol{\ell} = W_c \mathbf{z} + \mathbf{b}_c. \tag{4}$$

The upstream encoder supplies fixed patch features; attention aggregates a variable-size bag; and the linear classifier predicts its slide class. Softmax probabilities over classes are distinct from attention weights over patches.

B. Proposal A: Dual-Level Correction (Implemented)

Bag-level correction uses inverse-frequency training weights

$$w_c = \frac{N_{\text{train}}}{C\, n_c}, \tag{5}$$

where C is the number of classes and n_c is the training count for class c. Weighted cross-entropy is computed without reduction, then averaged across bags. This matters for batch size one: the standard weighted-mean reduction would cancel the single sample's class weight.

The sparse aggregator retains the highest $k = \max(1, \lceil r|\mathcal{V}|\rceil)$ attention logits and renormalizes softmax within the retained set. The primary fraction is $r = 0.10$. Patches outside that set contribute zero. Selection of the top-k support is discrete; retained scores and features remain differentiable. Hard selection can also discard useful context or amplify an early incorrect ranking.

TABLE II: Ablation variants. E0: Standard ABMIL; E1: Class-Weighted ABMIL; E2: Sparse-Attention ABMIL; E3: Combined ABMIL.

Variant	Class weighting	Top-10% attention
E0	No	No
E1	Yes	No
E2	No	Yes
E3	Yes	Yes

For clarity, E0 denotes *Standard ABMIL*, which uses all valid patches with ordinary cross-entropy. E1 denotes *Class-Weighted ABMIL*, which retains full attention but scales the supervised loss by the training class frequency. E2 denotes *Sparse-Attention ABMIL*, which uses ordinary cross-entropy and retains the top 10% of attention-scored instances. E3 denotes *Combined ABMIL*, which applies both changes. This naming is retained in the result tables to keep the full ablation identifiable.

The selected extension is Combined ABMIL (E3). E1 and E2 isolate its two components. Thus, E3 versus E0 measures the combined change, E1 versus E0 measures weighting alone, and E2 versus E0 measures sparse aggregation alone. Comparison of E3 against E1 and E2 tests whether the combination provides an additional benefit. All variants retain the same trainable parameter count, feature input, initialization protocol, and optimizer configuration. The 10% fraction is a fixed primary design choice rather than a demonstrated optimum; the separate validation-only fraction study examines sensitivity to that choice.

C. Proposal B: Bag-Composition Robustness (Proposed)

A separate direction examines whether slide predictions and attention distributions remain stable after random instance removal. A possible extension is consistency training across sampled views of the same bag. Evaluation would measure prediction flips, changes in probabilities, and attention agreement on shared patches. A limited evaluation probe retains 50% or 10% of test instances, with three sampling repeats. It measures random-removal stability, not the true tumor-patch fraction. Consistency training and quantitative localization are not implemented.

D. Implementation Verification

The implementation checks padded and unpadded attention, permutation invariance, selected-support size, gradient propagation, and checkpoint consistency. A read-only audit completed 472 checks, including replay of 26 original validation-selected checkpoints. The largest C16 training bag also passed a backward test on Apple MPS. The checks establish consistency between saved models and reported evaluation. They do not establish biological validity or absence of upstream biases.

V. Experimental Design

All primary models use the original course splits, seeds 42, 43, and 44, and 30 training epochs. AdamW uses learning rate 10^{-4} and weight decay 10^{-5}. Batch size is one, with no dropout. The starter's warmup configuration is inactive for every model, leaving a constant learning rate. Every model receives the same training budget. The checkpoint is the earliest epoch maximizing validation macro-AUROC plus macro-F1. Test predictions are evaluated only after checkpoint selection.

The primary comparison fixes $r = 10\%$ throughout. Accuracy, macro one-versus-rest (OVR) AUROC, macro-F1, and macro average precision (AP) are reported. Per-class recall, specificity, confusion matrices, and loss curves support interpretation. In C16, class-1 recall is sensitivity and class-0 recall is specificity.

Three-seed averages are reported with sample standard deviations. These seeds share one test cohort and are not independent patient samples. No statistical significance claim is made. The descriptive two-factor interaction in an outcome M is $M(E3) - M(E2) - M(E1) + M(E0)$; it is not a causal estimate of optimization mechanisms.

Following the initial test observations, attention fractions $r \in \{0.05, 0.10, 0.25, 0.50\}$ are evaluated using validation only. The fraction with the highest mean best-validation AUROC+F1 across seeds is selected separately by dataset and model. Ties prefer 10%, then 5%, 25%, and 50%. Saved selected checkpoints are evaluated without further updates. The existing 10% test results had already been observed, so the follow-up is exploratory even though the selection of new candidates does not use test performance.

C16 training prevalence is separately manipulated to 50%, 20%, and 5% positives, always using 140 distinct training slides. The corresponding positive counts are 70, 28, and 7. Each E0/E3 pair receives identical slide IDs. Validation and test sets remain fixed. Three paired model/sampling seeds are used: (42,2026), (43,2027), and (44,2028). Their variation combines optimization and training-cohort composition. Constant total sample size does not preserve positive-example diversity.

The supplementary C17 patient-grouped experiment is reported separately from the required course comparison. Group membership, class composition, and checkpoint selection all change with the new split. Therefore, the contrast does not isolate the causal effect of train–validation overlap.

The experiments were carried out on an Apple M5 Pro with unified memory of 24 GB, Python 3.14 and PyTorch 2.14.1 using MPS. An initial training-plus-validation epoch took 9.34, 9.43, and 10.16 s for C16, C17, and BRACS. A sequential runner preserved best and most recent model checkpoints, optimizer states, random-number generator states, per-run specifications, dataset IDs, training histories, and prediction files. All 129 queued tasks completed, comprising 120 training runs, six selection stages, and three instance-subsampling evaluations. Additional test-only releases were not separate training runs or independent test populations. These execution details support reproducibility, but do not replace external

validation.

VI. Results

E3 reduces mean accuracy relative to E0 by 3.36, 11.33, and 6.59 percentage points on C16, C17, and BRACS, respectively (Table III; Fig. 1). E2 improves mean accuracy slightly on C16 and by 5.17 points on course-split C17, but performs worse on BRACS. Class weighting alone does not consistently improve accuracy. BRACS E1 improves macro-F1 despite slightly lower accuracy.

E3 increases mean AUROC on C16 and C17 while reducing accuracy. AUROC measures ranking across thresholds; accuracy uses the fixed argmax decision. A ranking improvement need not translate to better class decisions. This discrepancy alone does not prove miscalibration or justify test-tuned thresholds.

The mean descriptive accuracy interaction is -2.58, -5.50, and -1.55 percentage points for C16, C17, and BRACS. These values do not demonstrate a specific biological or optimization mechanism.

C16 validation scores frequently saturate early, selecting early checkpoints. For BRACS E0, final training CE falls to 0.079–0.113 while validation CE rises to 2.84–3.06, compared with 1.37–1.41 at the selected epochs. This pattern is consistent with late overfitting or overconfidence. Reported test scores use validation-selected checkpoints, not the final-epoch weights. Weighted training loss and unweighted validation CE represent different objectives and should not be interpreted as a direct generalization gap.

On C17, E3 predicts class 1 for 24.00% of test slides, versus 6.17% for E0; the true test prevalence is 8%. Meanwhile, the class-0 prediction share falls from 59.00% to 45.50%. Changes in minority-class recall can therefore coexist with substantially more majority-class errors. These observations are compatible with altered decision frequencies but do not identify class weighting as the sole cause.

At 50% training prevalence, E3 has slightly higher accuracy and sensitivity than E0. At 20%, mean accuracy is similar. The largest accuracy separation occurs when only 5% of training slides are positive. Mean accuracy is 78.29% for E0 and 71.58% for E3; mean sensitivity is 42.86% and 40.82%, respectively. The E3 accuracy SD reaches 11.51 points. The fixed sample count removes variation in total training-bag count. However, it does not isolate class imbalance: the number of distinct positive training slides falls from 70 to 7 and negative-sample composition also changes. Sample-seed and model-seed effects are not separately estimated. These data do not establish robust behavior for all low-prevalence cohorts. This experiment changes training-bag prevalence, not the true proportion of tumor patches within bags.

C16 selects 10% for both sparse variants, and C17 selects 10% for E2 but 5% for E3. BRACS selects 25% for both. The selected fractions do not restore a consistent combined-method advantage. C16 validation scores saturate across multiple fractions, making the fixed tie rule consequential. The earlier use of the same test cohort limits confirmatory interpretation.

A separate instance-subsampling probe retains 50% and 10% of test patches for seed-42 E0 and E3, with three fixed random repeats. The associated prediction-flip and probability-change values are preserved in the repository. Random removal does not control tumor content and is not evidence of localization accuracy.

E2 exceeds E0 in mean accuracy on the course split (82.50% versus 77.33%), but not on the patient-independent protocol (77.83% versus 79.83%). E3 remains below E0 in both. This difference cannot be attributed solely to patient overlap because the training cohort, validation class composition, and selected epochs also change. The grouped validation set contains only one class-1 slide.

The two official course slides and the tumor XML file were processed with the provided WSI visualization routine. Coordinates are interpreted at level zero. A 512-pixel patch footprint is assumed because extraction metadata does not verify the patch size. Overlay colors represent within-slide attention percentiles, not calibrated tumor probabilities. These qualitative examples do not measure localization performance.

VII. Discussion

A. Evidence for the Three Hypotheses

H1: Bag-level class weighting. Class-weighted ABMIL (E1) does not consistently improve the measured outcomes. Its mean accuracy is below the baseline on all three course datasets. BRACS shows a different trade-off: macro-F1 rises from 0.390 to 0.412 although accuracy declines slightly. This indicates that a class-sensitive metric can improve without a corresponding increase in overall correct predictions. On C17, weighting raises class-1 recall in some settings but substantially reduces class-0 recall. The results therefore do not support a general accuracy benefit from inverse-frequency weighting. They also do not establish that weighting is ineffective for every class or metric.

H2: Sparse instance aggregation. Sparse-Attention ABMIL (E2) increases mean accuracy on course-split C16 by 0.26 percentage points and on C17 by 5.17 points. Its mean C16 AUROC also increases from 0.932 to 0.961. In contrast, BRACS accuracy declines by 3.88 points. The C17 accuracy advantage disappears when a patient-independent train/validation protocol is used. This dependence on dataset and split prevents a general claim of superior sparse-evidence robustness. More fundamentally, no verified patch-level tumor labels are available for these full bags. The experiments measure slide classification under a sparse pooling rule, not sensitivity as a function of the true tumor-patch ratio.

H3: Interaction of class weighting and sparse attention. Combined ABMIL (E3) fails to improve mean accuracy over Standard ABMIL on any of the three original

TABLE III: Primary course-split test results at fixed 10% sparse attention (mean ± sample SD; three seeds). Accuracy is reported in percent; other metrics are in $[0, 1]$.

Dataset	Variant	Accuracy (%)	Macro OVR AUROC	Macro-F1	Macro-AP
C16	E0	90.44 ± 3.58	0.932 ± 0.031	0.895 ± 0.039	0.935 ± 0.031
	E1	89.41 ± 5.44	0.927 ± 0.034	0.886 ± 0.056	0.931 ± 0.034
	E2	90.70 ± 3.88	0.961 ± 0.010	0.896 ± 0.046	0.964 ± 0.008
	E3	87.08 ± 4.27	0.959 ± 0.011	0.860 ± 0.043	0.962 ± 0.008
C17	E0	77.33 ± 3.25	0.838 ± 0.023	0.559 ± 0.053	0.580 ± 0.019
	E1	66.33 ± 6.71	0.813 ± 0.006	0.499 ± 0.042	0.543 ± 0.019
	E2	82.50 ± 1.50	0.855 ± 0.022	0.564 ± 0.042	0.617 ± 0.034
	E3	66.00 ± 8.00	0.846 ± 0.014	0.540 ± 0.025	0.616 ± 0.021
BRACS	E0	46.90 ± 0.67	0.804 ± 0.016	0.390 ± 0.021	0.439 ± 0.015
	E1	45.74 ± 2.42	0.802 ± 0.013	0.412 ± 0.021	0.448 ± 0.014
	E2	43.02 ± 2.33	0.788 ± 0.007	0.359 ± 0.030	0.426 ± 0.003
	E3	40.31 ± 2.42	0.782 ± 0.020	0.349 ± 0.041	0.415 ± 0.034

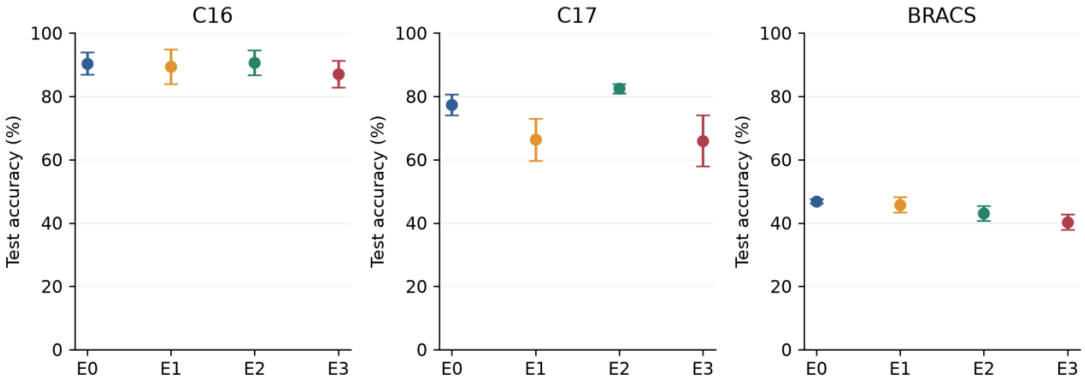


Fig. 1: Primary accuracy by variant on the course splits. Error bars show sample SD across three model seeds, not uncertainty over independent patient cohorts.

course datasets. The mean declines are 3.36, 11.33, and 6.59 percentage points on C16, C17, and BRACS. Mean macro-F1 also declines in all three. The two components are therefore not complementary under the tested configuration. The descriptive interaction, $M(E3) - M(E2) - M(E1) + M(E0)$, is negative for course-split accuracy, but differs by metric and protocol. The grouped C17 interaction is approximately zero. No stable mechanistic interaction can be inferred from those outcome differences alone.

B. Why Ranking and Decisions Can Disagree

AUROC evaluates whether positive and negative examples tend to receive the correct relative scores across thresholds. Accuracy evaluates the final argmax class decision. As a result, ranking quality can increase even when the predicted class frequencies shift in an unfavorable direction. For C16, E3 has higher mean AUROC than E0 (0.959 versus 0.932) but lower mean accuracy (87.08% versus 90.44%). A similar pattern occurs on C17. The class-wise results help explain the decision trade-off: E3 predicts C17 class 1 for 24.00% of test slides on average, although the observed prevalence is 8%. At the same time, the predicted class-0 share falls from 59.00% to 45.50%.

These differences are compatible with changes in decision distributions caused by the training objective, sparse selection, or both. They do not prove that predicted probabilities are miscalibrated. Calibration diagnostics, negative log-likelihood, and Brier scores remain supplementary observations. The original argmax decision rule is retained; test-set outcomes are not used to choose a more favorable threshold.

C. Possible Sources of Combined-Method Degradation

Inverse-frequency weighting is a conventional class-rebalancing strategy [3]. It multiplies the gradient contribution of each bag according to its class. It can make uncommon classes more influential, but the weights do not increase the number of unique minority-class patients or morphologies. When a class has very few slides, a larger weight can increase sensitivity to the particular examples observed during training. Hard top-k attention introduces a separate restriction: patches outside the selected support cannot influence the pooled representation during that forward pass. If early attention rankings emphasize a confounder, the model may reinforce an incomplete or misleading subset. Discarding low-scoring patches may also remove context that contributes to histological classifi-

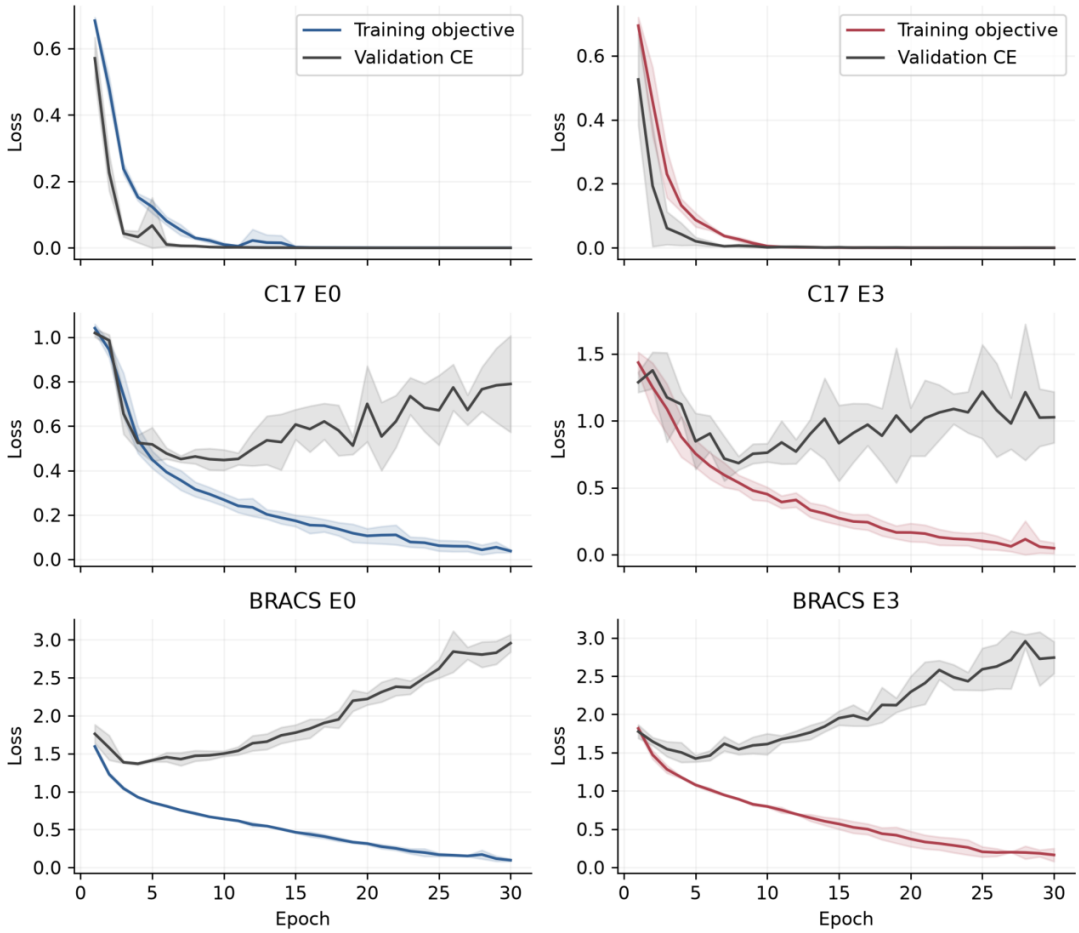


Fig. 2: Training objective and unweighted validation cross-entropy over 30 epochs, for E0 and E3. Shading indicates variation across seeds. The weighted E3 training objective is not directly comparable with unweighted validation loss.

cation. Unlike differentiable sparsemax [11], hard top-k membership is discrete, although selected scores remain differentiable.

A combined model may therefore amplify unstable instance selection while giving higher gradient weight to rare classes. This is a plausible explanation for E3's poorer outcomes, not a demonstrated causal mechanism. The current study does not trace attention against verified tumor labels or isolate the optimization effect through controlled perturbations. Furthermore, the fraction study does not rescue E3 through validation selection: the selected E3 fractions are 10% for C16, 5% for C17, and 25% for BRACS. All selected configurations remain exploratory because the diagnostic study was designed after initial test results had been observed.

D. Training Scarcity and Protocol Sensitivity

The controlled C16 prevalence experiment shows increasing classification difficulty as the proportion of positive training slides decreases. At 5% prevalence, only seven distinct positive training bags remain. Mean E0 and E3 accuracies are 78.29% and 71.58%, respectively, with substantial variability in E3. This result does not isolate the effect of class prevalence alone. Positive-example diversity is reduced at the same time, and the three repeats vary both model initialization and sample composition. The experiment supports a narrower conclusion: the combined method does not demonstrate reliable protection under the tested low-positive-data regime.

The C17 split audit exposes another source of uncertainty. Twenty-seven patients overlap between the original training and validation partitions, although the course test set remains patient-disjoint. The supplementary grouped protocol removes this overlap, but changes training membership, validation class composition, and selected checkpoints. Its validation set includes only one class-1 slide. Thus, the difference between grouped and course results cannot be attributed solely to leakage. It nevertheless demonstrates that the apparent benefit of sparse attention on C17 is sensitive to the model-selection protocol.

E. Limitations

Several limitations define the scope of the present findings. First, all experiments use fixed, pre-extracted

TABLE IV: Per-class test recall (%, mean ± sample SD across three seeds). Numeric labels follow the supplied datasets; unsupported semantic mappings are omitted.

Dataset	Variant	Class 0	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
C16	E0	96.67 ± 3.15	80.27 ± 5.14	–	–	–	–	–
	E1	94.17 ± 7.11	81.63 ± 3.53	–	–	–	–	–
	E2	98.75 ± 1.25	77.55 ± 9.35	–	–	–	–	–
	E3	91.67 ± 10.18	79.59 ± 10.80	–	–	–	–	–
C17	E0	88.15 ± 4.77	18.75 ± 6.25	31.48 ± 11.56	87.41 ± 3.39	–	–	–
	E1	68.04 ± 12.18	31.25 ± 6.25	18.52 ± 8.49	93.33 ± 3.85	–	–	–
	E2	96.69 ± 0.83	12.50 ± 6.25	22.22 ± 5.56	93.33 ± 2.22	–	–	–
	E3	64.46 ± 16.96	29.17 ± 25.26	29.63 ± 8.49	97.78 ± 0.00	–	–	–
BRACS	E0	50.00 ± 16.67	45.83 ± 9.55	25.93 ± 12.83	3.03 ± 5.25	27.78 ± 17.35	52.78 ± 29.27	88.33 ± 12.58
	E1	66.67 ± 0.00	10.42 ± 13.01	29.63 ± 16.97	42.42 ± 13.89	33.33 ± 14.43	50.00 ± 30.05	81.67 ± 11.55
	E2	33.33 ± 28.87	50.00 ± 10.83	25.93 ± 6.42	9.09 ± 9.09	13.89 ± 12.73	50.00 ± 8.33	80.00 ± 8.66
	E3	44.44 ± 19.25	14.58 ± 9.55	25.93 ± 12.83	24.24 ± 27.77	41.67 ± 16.67	25.00 ± 8.33	83.33 ± 10.41

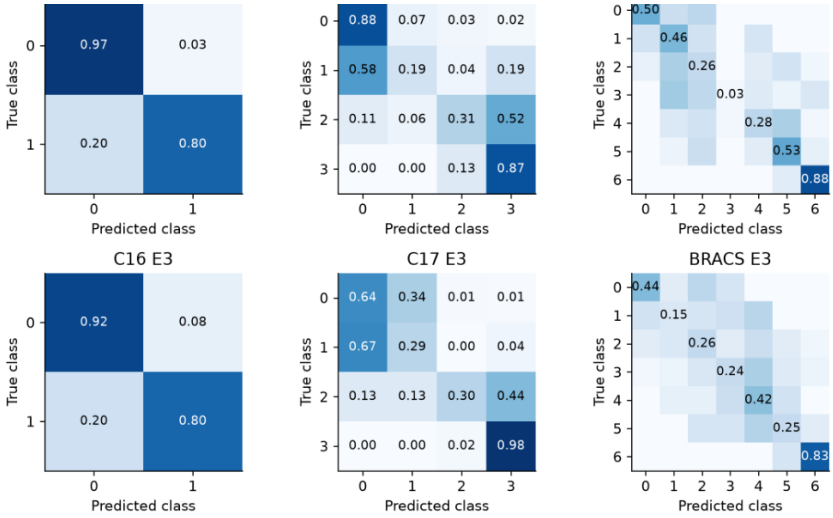


Fig. 3: Mean row-normalized confusion matrices on the course test sets. E0 is shown above E3. All rows use the same class order as the data.

TABLE V: Controlled C16 training prevalence (mean ± sample SD across three paired model/sampling repeats). Every condition uses 140 distinct training slides. Validation and test data are held fixed.

Positive bags	Unique positives	Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	Macro OVR AUROC
50%	70	E0	88.37 ± 2.69	76.19 ± 6.23	95.83 ± 2.89	0.932 ± 0.023
		E3	89.92 ± 2.05	80.95 ± 4.25	95.42 ± 5.77	0.944 ± 0.020
20%	28	E0	85.27 ± 1.55	63.27 ± 5.40	98.75 ± 1.25	0.879 ± 0.040
		E3	84.75 ± 2.72	65.99 ± 7.17	96.25 ± 0.00	0.888 ± 0.030
5%	7	E0	78.29 ± 4.65	42.86 ± 12.24	100.00 ± 0.00	0.831 ± 0.030
		E3	71.58 ± 11.51	40.82 ± 17.79	90.42 ± 11.61	0.737 ± 0.168

patch features. This design enables controlled comparisons of attention aggregation and loss weighting without introducing differences in feature extraction. However, it does not establish whether the same trends would hold with alternative feature encoders or end-to-end feature adaptation. The conclusions therefore primarily concern MIL aggregation and optimization under the evaluated feature representations.

Second, the experiments evaluate two relatively simple imbalance-handling strategies: inverse-frequency class weighting and hard top-k attention. Although these choices make the individual contributions easy to isolate, they do not cover the broader range of possible weighting functions, instance-selection mechanisms, or adaptive aggregation methods. The primary comparison uses a fixed attention-retention fraction of 10%, while a separate sensitivity analysis examines additional fractions. More flexible instance-selection strategies may behave differently when informative evidence is distributed across multiple regions.

Third, the evaluation uses three random seeds and fixed test cohorts. These repetitions provide an initial assessment of training variability but cannot fully characterize uncer-

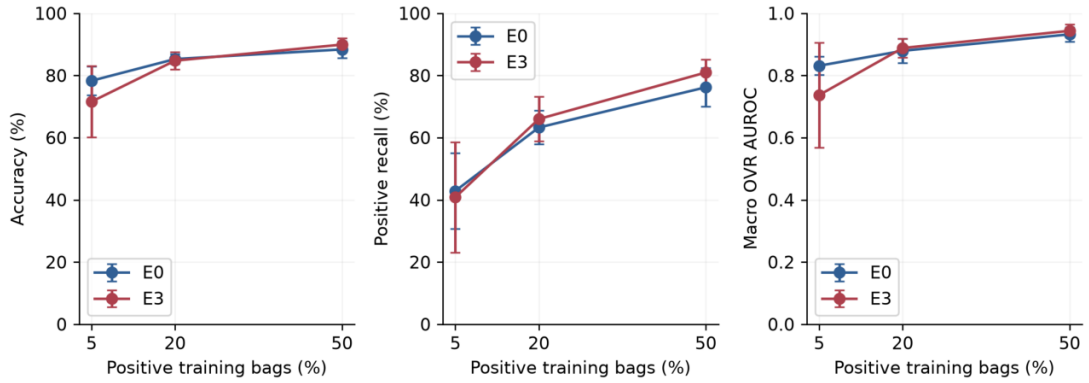


Fig. 4: Accuracy, positive recall, and macro OVR AUROC versus training-positive prevalence. Error bars combine model initialization and sampled-cohort variation.

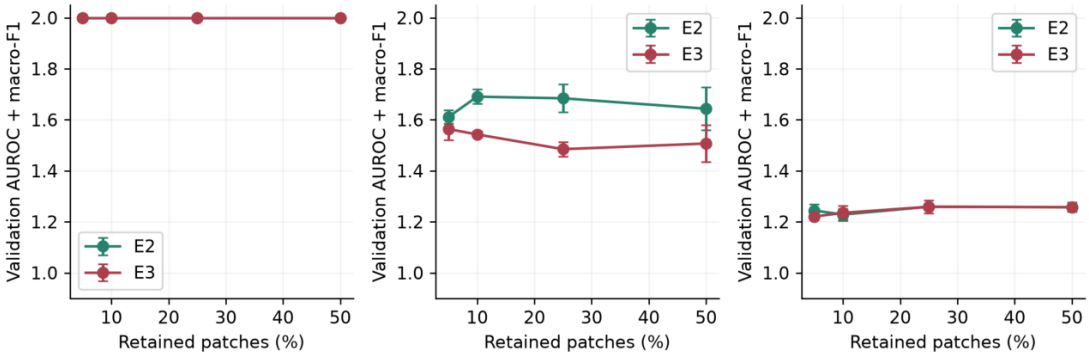


Fig. 5: Mean best-validation AUROC+macro-F1 over the prespecified fraction grid. No newly trained candidate is selected using test scores.

TABLE VI: Validation-selected attention fractions and corresponding held-out outcomes (three model seeds). These exploratory results do not replace the fixed-10% primary comparison.

Dataset	Model	Fraction	Mean validation AUROC+F1	Accuracy (%)	Macro OVR AUROC	Macro-F1
C16	E2	10%	2.0000	90.70 ± 3.88	0.961 ± 0.010	0.896 ± 0.046
C16	E3	10%	2.0000	87.08 ± 4.27	0.959 ± 0.011	0.860 ± 0.043
C17	E2	10%	1.6906	82.50 ± 1.50	0.855 ± 0.022	0.564 ± 0.042
C17	E3	5%	1.5633	61.67 ± 11.73	0.844 ± 0.014	0.522 ± 0.042
BRACS	E2	25%	1.2590	40.70 ± 2.33	0.800 ± 0.004	0.335 ± 0.025
BRACS	E3	25%	1.2590	38.37 ± 4.03	0.790 ± 0.013	0.341 ± 0.024

tainty across different patient samples or dataset partitions. The prevalence experiments also involve changes in the sampled training cases. Consequently, variability arising from data selection and model optimization cannot be completely separated. The C17 grouped analysis further illustrates the importance of patient-level independence when interpreting differences between methods.

Finally, the available slide-level labels do not identify which individual patches contain malignant tissue. Attention maps can illustrate which image regions influence aggregation, but they cannot independently establish localization accuracy. The WSI visualizations should therefore be interpreted as qualitative examples of model behavior rather than quantitative evidence of tumor localization. These limitations motivate additional experiments with region-level annotations and more diverse evaluation settings.

F. Future Directions

The results suggest several opportunities for improving imbalance-aware MIL beyond the strategies evaluated in this study.

One direction is to investigate adaptive attention sparsity. Hard top-k aggregation retains a predetermined fraction of instances for every slide, regardless of its tissue composition or evidence distribution. A fixed threshold may be restrictive when the number of informative patches varies substantially across slides. Future methods could learn the

TABLE VII: C17 performance using original course or patient-independent train/validation partitions on the same center-held-out test set (mean ± sample SD across three model seeds).

Protocol	Model	Accuracy (%)	Macro OVR AUROC	Macro-F1
Course	E0	77.33 ± 3.25	0.838 ± 0.023	0.559 ± 0.053
	E1	66.33 ± 6.71	0.813 ± 0.006	0.499 ± 0.042
	E2	82.50 ± 1.50	0.855 ± 0.022	0.564 ± 0.042
	E3	66.00 ± 8.00	0.846 ± 0.014	0.540 ± 0.025
Patient-independent	E0	79.83 ± 2.47	0.836 ± 0.016	0.524 ± 0.011
	E1	67.33 ± 7.23	0.809 ± 0.017	0.483 ± 0.028
	E2	77.83 ± 1.15	0.827 ± 0.014	0.512 ± 0.035
	E3	65.33 ± 17.19	0.803 ± 0.005	0.505 ± 0.007

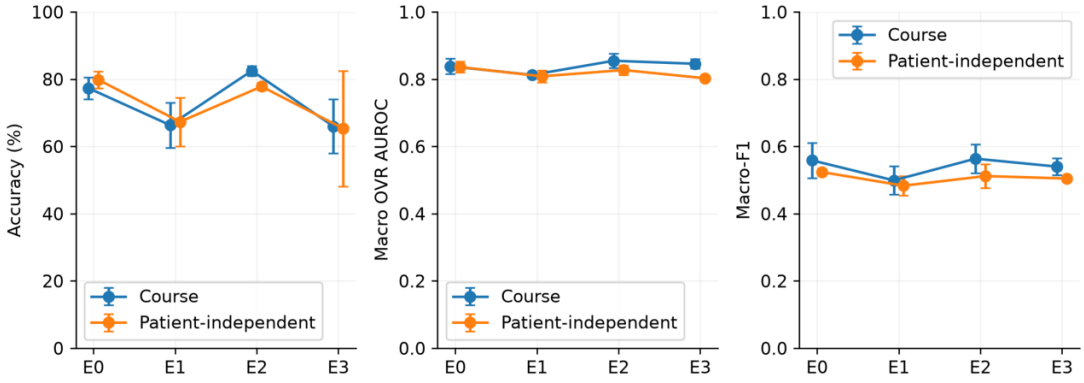


Fig. 6: Comparison of course and patient-independent C17 protocols. The test cohort is unchanged. Train/validation composition and checkpoint selection differ.

retention fraction, apply soft rather than hard selection, or combine high-attention instances with broader contextual information. Such approaches could preserve the benefits of selective aggregation while reducing the risk of discarding relevant evidence.

A second direction concerns more targeted approaches to class imbalance. Inverse-frequency weighting increases the contribution of underrepresented classes, but it does not explicitly account for differences in class difficulty, prediction confidence, or the availability of discriminative instances. Alternative objectives, including focal or class-balanced losses, could be compared with sampling-based strategies under matched training conditions. Evaluation should consider macro-averaged metrics, per-class sensitivity, and probability calibration alongside overall accuracy. This would help distinguish improvements in minority-class recognition from changes in the model's overall prediction distribution.

A third direction is to examine the interaction between representation learning and MIL aggregation. The current study isolates the aggregation stage using fixed patch features. Future experiments could evaluate whether pathology-specific encoders, self-supervised representations, or limited encoder fine-tuning improve the identification of informative patches. In particular, stronger representations may change which instances receive high attention and whether sparse aggregation is beneficial. This would help determine whether the observed limitations arise primarily from instance selection or from the information available in the patch features.

Finally, future work could incorporate a small amount of region-level supervision and broader patient-independent evaluation. Expert-annotated regions would allow attention concentration to be compared against actual lesion locations rather than assessed only through slide-level predictions. Experiments across additional patient splits and external cohorts could also establish whether the observed differences generalize beyond the current evaluation settings. Together, these extensions would support a more detailed understanding of how class imbalance, feature quality, and evidence selection affect weakly supervised pathology models.

VIII. Conclusion

This study investigated whether addressing class imbalance at both the bag and instance levels improves attention-based multiple instance learning for whole-slide histopathology classification. Two modifications to standard ABMIL were examined: inverse-frequency class weighting to address unequal class representation across training slides, and sparse attention aggregation to emphasize a smaller subset of potentially informative patches. Their individual and combined effects were evaluated on CAMELYON16, CAMELYON17, and BRACS through controlled ablations and additional robustness analyses.

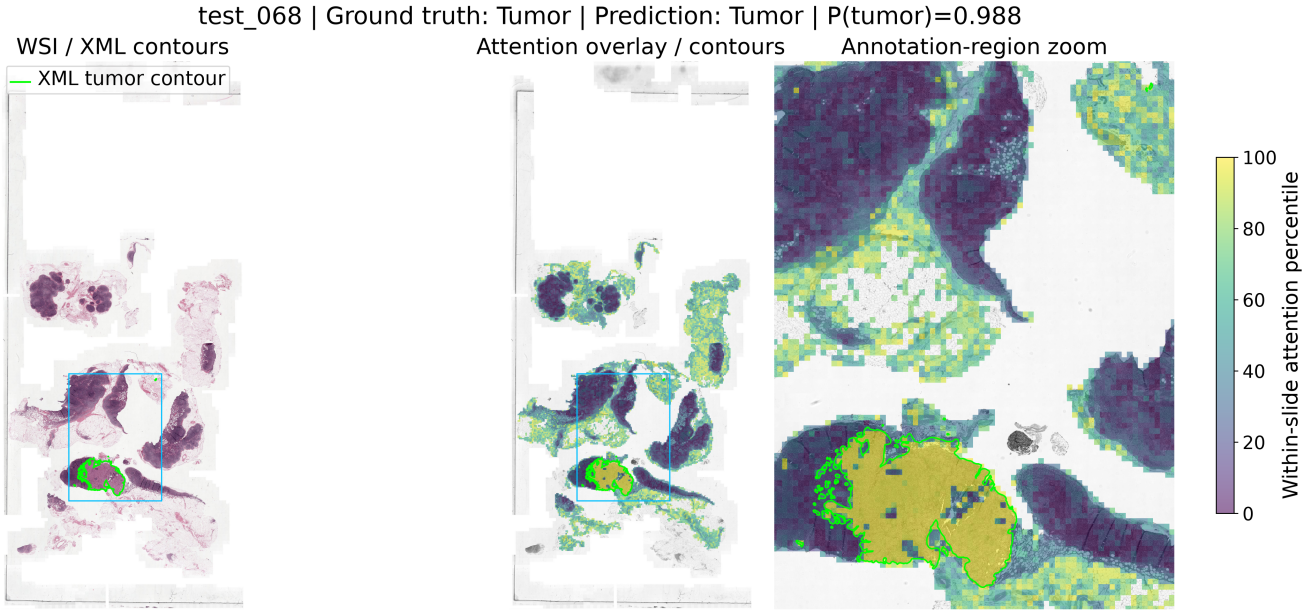


Fig. 7: CAMELYON16 tumor slide `test_068`, using the E0 seed-42 checkpoint selected at epoch 3. The figure shows the WSI with XML contours, an attention overlay, and an annotation-region zoom. The slide contains 15,037 patches and 45 XML contours; its predicted tumor probability is 0.988.

The findings show that the benefits of these modifications are not consistent across datasets. Sparse attention improved mean classification accuracy on the original CAMELYON16 and CAMELYON17 evaluation splits but reduced performance on BRACS. Class weighting did not consistently improve overall accuracy, while the combined approach underperformed standard ABMIL across all three datasets. Additional experiments involving reduced positive-class prevalence, attention-retention fractions, and patient-independent validation further demonstrated that the observed effects depend on the data distribution and evaluation protocol.

These results highlight an important distinction between the motivation for an architectural modification and its empirical effectiveness. Although class weighting and sparse aggregation target different aspects of the learning problem, their combination does not necessarily produce complementary benefits. Emphasizing underrepresented classes changes the optimization objective, while restricting attention changes the evidence available for slide-level prediction. Their interaction therefore requires empirical validation rather than an assumption of additive improvement.

More broadly, the study emphasizes that reliable weakly supervised histopathology classification depends on more than the choice of attention mechanism. The balance between global tissue context and localized discriminative evidence, the quality of patch representations, and the independence of evaluation samples can all influence the interpretation of model performance. Future progress may therefore benefit from adaptive instance selection, improved representation learning, and evaluation protocols that jointly assess class-wise classification and region-level evidence.

Ultimately, the contribution of this study lies not in demonstrating a universally superior ABMIL variant, but in identifying conditions under which intuitive imbalance-aware modifications succeed or fail. Such controlled investigations provide a useful foundation for developing more robust, interpretable, and data-efficient MIL systems for computational pathology.

Code and Reproducibility

The Project A repository is intended to contain the completed notebook, PyTorch implementation, experiment configurations, numeric results, figures, and reproducibility instructions. Separate report and code reproducibility checklists document the study protocol and executable artifacts, respectively. Source data are retrieved from the official course feature repository [13]; large HDF5 and WSI files are not embedded in the code archive. The experiment manifest records configurations, model seeds, training histories, split IDs, metrics, and selected checkpoints. Tests cover attention normalization, masking, gradient flow, checkpoint recovery, and evaluation consistency. MPS execution is not guaranteed to be bitwise deterministic across systems.

Project repository:
`https://github.com/anniekang1112/ECE1512-Project-A`.

A. Reproducibility Scope

The repository separates code from the written report. It provides implementation files, experiment configurations,

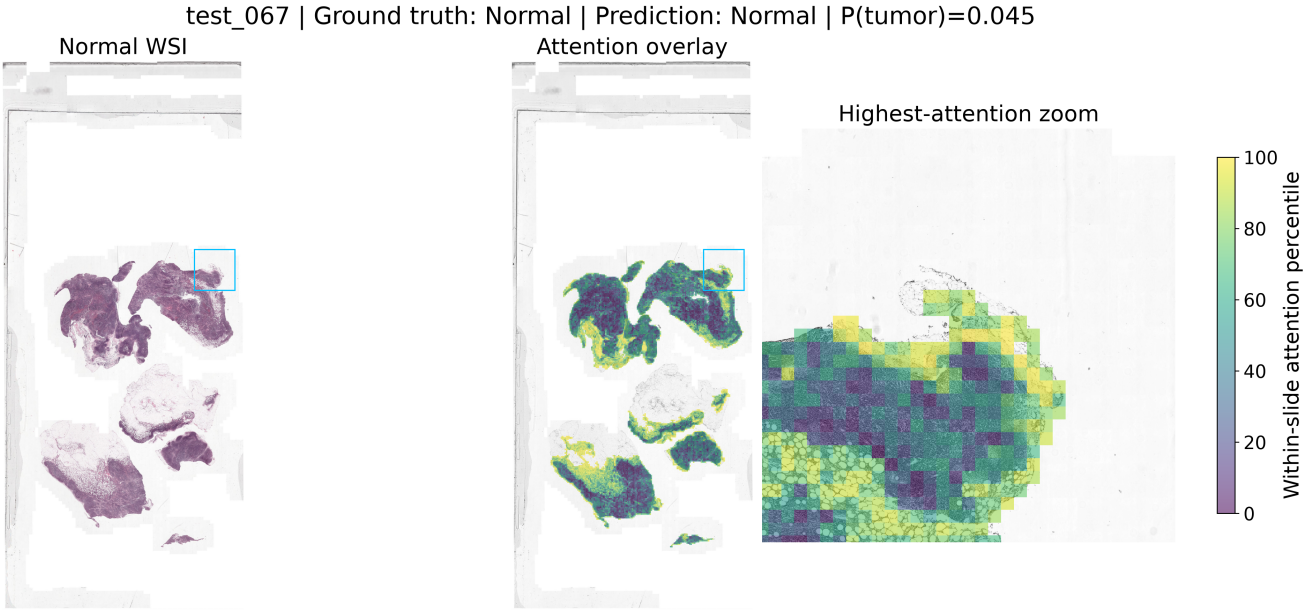


Fig. 8: CAMELYON16 normal slide `test_067`, using the same checkpoint. The final panel centers on the highest-attention patch. The slide contains 11,926 patches and has predicted tumor probability 0.045.

per-seed results, figures, and a README with commands and data-access links. Large course-provided HDF5 and WSI assets are omitted from the repository bundle; their locations and integrity information are documented. Reproduction of the original attention overlays requires the designated slides, XML annotation, and saved weights. No unpublished result is inferred from an absent checkpoint.

TABLE VIII: Supplementary Appendix A: Primary course-split experiments by seed. Accuracy is in percent. AUROC, F1 and AP are in $[0, 1]$. Selected epochs maximize validation AUROC+F1.

Dataset	Model	Seed	Accuracy	AUROC	Macro-F1	Macro-AP	Epoch
C16	E0	42	88.37	.9224	.8714	.9282	3
C16	E0	43	88.37	.9064	.8739	.9086	3
C16	E0	44	94.57	.9666	.9406	.9697	3
C16	E1	42	89.92	.9212	.8896	.9274	3
C16	E1	43	83.72	.8957	.8279	.8991	3
C16	E1	44	94.57	.9635	.9406	.9673	3
C16	E2	42	86.82	.9730	.8490	.9727	2
C16	E2	43	90.70	.9569	.8976	.9599	2
C16	E2	44	94.57	.9536	.9406	.9595	3
C16	E3	42	86.82	.9712	.8490	.9710	2
C16	E3	43	91.47	.9546	.9075	.9580	2
C16	E3	44	82.95	.9505	.8248	.9559	4
C17	E0	42	77.50	.8225	.5332	.5596	21
C17	E0	43	80.50	.8643	.6191	.5961	19
C17	E0	44	74.00	.8280	.5232	.5854	18
C17	E1	42	61.50	.8113	.4817	.5293	28
C17	E1	43	63.50	.8081	.4686	.5343	30
C17	E1	44	74.00	.8191	.5470	.5650	30
C17	E2	42	81.00	.8307	.5160	.5813	12
C17	E2	43	82.50	.8617	.5783	.6208	12
C17	E2	44	84.00	.8720	.5966	.6486	9
C17	E3	42	66.00	.8396	.5686	.6192	5
C17	E3	43	58.00	.8619	.5313	.6351	9
C17	E3	44	74.00	.8362	.5202	.5928	12
BRACS	E0	42	46.51	.7855	.3716	.4217	3
BRACS	E0	43	46.51	.8132	.3851	.4450	7
BRACS	E0	44	47.67	.8122	.4127	.4504	8
BRACS	E1	42	47.67	.7873	.4177	.4358	5
BRACS	E1	43	46.51	.8041	.4301	.4461	5
BRACS	E1	44	43.02	.8137	.3886	.4635	6
BRACS	E2	42	45.35	.7913	.3829	.4237	12
BRACS	E2	43	43.02	.7939	.3686	.4296	10
BRACS	E2	44	40.70	.7802	.3253	.4244	11
BRACS	E3	42	39.53	.7605	.3163	.3762	3
BRACS	E3	43	43.02	.7888	.3949	.4370	12
BRACS	E3	44	38.37	.7982	.3345	.4325	10

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